Yiyu Zhu

Full-Stack Software Engineer

zhuyy8612@gmail.com · GitHub portfolio · Remote contracts · 20 hours/week · China UTC+8

Builds Node.js and Python applications with persistent state, failure recovery, and reproducible regression tests. Works on AI application integration at China Telecom.

RunLedger | Durable Webhook Delivery

Independent project · Node.js 24 · SQLite · Browser console · Nov 2026 · Source and test steps

Delivers AI job completion notifications and lets an operator trace attempts, recover from receiver outages, and replay dead-letter deliveries from a browser.

- Writes the event and delivery in one SQLite transaction. A repeated idempotency key returns the existing delivery for the same canonical payload and HTTP 409 for a changed payload.
- Claims jobs with expiring leases and fences stale worker writes with a lease token. Retries network errors, 429s, and 5xx responses with backoff. Manual replay retains prior attempts and the original delivery ID.
- Signs the stable delivery ID, timestamp, and raw message with HMAC. A persistent receiver inbox records each ID once under at-least-once delivery.
- Ran a two-process claim race and killed the winning process. A new lease reclaimed the same delivery as attempt two. A local HTTP fault test reached dead letter after 503s and returned 200 on replay with its history intact.
- Sent 200 requests in local batches of 20 concurrent submissions with two workers. Deduplicated 100 repeat submissions and verified 100 unique deliveries and 100 receiver effects.

Selected Projects

OrderCheck | Order Data Quality Workbench Validates amounts with Decimal and quarantines conflicting groups by business key. A seven-sheet Excel report records source rows, rules, and input hashes. A 51-test suite covers validation, export, and the shared Streamlit and CLI logic.

API Contract Lab | HTTP and SSE Regression Exercises malformed JSON, missing SSE termination, and timeouts against a local HTTP server. Eighteen self-tests cover contract and retry behavior, with HTML diagnostics and JSON output.

LeadReview | AI Extraction and Human Review Scores model-extracted information with deterministic rules and separates proposed actions from approved actions. A JavaScript source module generates the importable n8n workflow, backed by 17 assertions.

Experience

China Telecom | AI Application Engineering and Systems Integration · Jun 2026–Present

- Reproduce integration failures across input collection, configuration, recognition output, and event delivery. Adjust test conditions and retest to isolate the cause.
- Document expected and actual behavior, reproduction steps, and retest results. Maintain configuration checks for integration and regression testing.

Education and Languages

City University of Hong Kong · MSc, Electronic Information Engineering · Sep 2025–Jun 2026

Sun Yat-sen University · BEng, Aerospace Engineering · Sep 2021–Jun 2025

Native Mandarin · English CET-6 505 · English-medium postgraduate study